Response to the Reviewers' Comments:
Truly Mobile Sub-Terahertz Communications Beyond 6G via Just-in-Time IMU-Assisted Beam Management
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Dear Editor-in-Chief Prof. Alouini, Dear Reviewers,

Thank you very much for sharing your expert opinions on our work. We have carefully read every comment and made the corresponding clarifications and revisions; a detailed point-by-point response follows. The major changes in this revision are summarised below for quick reference:

1. **Related-work positioning** against six families of prior work (Table 1).
2. **Three new parameter-sensitivity figures** covering four sweeps: γ (HS), β (IMU-Assist), JIT η, and T_{IMU}.
3. **New *Trigger Boundary and Edge of Coverage* subsubsection** in *Methods/JIT Beam Realignment* (geometry, motion adaptation, UE-side detection, RF-side fallback).
4. **IMU drift analysis.**

In the revised manuscript, blue font indicates revised, added, or otherwise updated text. The marked-up manuscript is provided as `final_revision3.pdf` and the clean version as `final_3.pdf`.

Editor

Clearly differentiate your pre-failure trigger and coordinated AP/UE mechanism from prior predictive tracking and IMU-assisted beam management. Provide a more detailed related-work comparison beyond the two baselines.

We have expanded the related-work positioning in the revised manuscript. The two reactive baselines (Hierarchical Search and IMU-Assist) remain the most directly comparable alternatives at the same protocol-level scope as JIT; JIT is now also contrasted against six families of prior work, summarized in Table 1.

What is new in JIT relative to these families is the combination of UE-side IMU motion sensing, a pre-failure trigger that anticipates the next beam-edge crossing, and coordinated AP/UE realignment via a single in-band sub-THz control frame. To our knowledge, no prior work combines all three within a single sub-THz protocol with hardware-in-the-loop validation. 3GPP NR Beam Failure Recovery [1, 2] sends a discrete PRACH-based recovery message only after the MAC entity has counted N consecutive Beam Failure Instances at L1; RF-based proactive failure prediction [3, 4] forecasts a failure from RF time-series or sub-6 GHz features and triggers a standard beam switch, but uses no motion sensing and cannot anticipate rotation-induced edge crossings; AP-side multimodal sensing [5, 6, 7] requires LiDAR, radar, vision, or GPS hardware at every AP; UE-side orientation-based work [8, 9] uses orientation primarily for codebook selection rather than to anticipate the next edge crossing; continuous IMU streaming [10] exchanges IMU data on every tracking interval over a separate out-of-band control channel; and bilateral pilot-driven coordination [11, 12, 2] incurs overhead regardless of whether the link is at risk. JIT sends one short frame only when a crossing is imminent, for an overhead of less than 0.4%.

Table 1: Beam-management work families positioned against JIT.

Approach (refs.)	Sensor / Modality	Trigger timing	AP+UE coord.	Trigger source
3GPP NR BFR [1, 2]	RF only	Post-failure	UE→AP	RF degradation
3GPP Rel-18 BM-Case [11, 12]	RF pilots / ref. signals	Pre-failure	Bilateral pilots	Periodic pilot sweep
RF-side pre-failure prediction [3, 4]	RF (CSI / sub-6 GHz)	Pre-failure	Unilateral (UE)	RF time-series model
Multimodal AP-side sensing [5, 6, 7]	LiDAR / camera / GPS / radar	Pre-failure	Unilateral (AP)	External sensor fusion
UE-orientation / VR head pose [8, 9]	IMU orient. (one-shot)	Pre-failure	Unilateral (UE)	Orientation → codebook
Continuous IMU streaming [10]	IMU + out-of-band link	Pre-failure	Bilateral	Periodic IMU samples
JIT (this work)	**IMU + RF**	**Pre-failure**	**UE→AP single frame**	**IMU edge-cross. prediction**

Show how JIT's performance varies with key parameters and verify that baselines are fully tuned. Without this, it is unclear whether gains come from the core mechanism or parameter choices.

We have added two parameter-tuning figures and one timescale-robustness figure, each evaluated under both translational and rotational mobility: Fig. 1 sweeps the spiral growth factor γ (Hierarchical Search) and the refinement step factor β (IMU-Assist); Fig. 2 sweeps JIT's guard fraction η; Fig. 3 sweeps the IMU sampling period T_{IMU}. To rule out a cherry-picked operating point, each parameter sweep uses the same translational (0.5–20 m/s^2) and rotational (0.5–700 deg/s^2) peak acceleration grids used throughout the main paper, with the same $n = 5{,}000$ Monte-Carlo runs per data point. The full acceleration sweep is drawn as faint background lines; the min, typical (3 m/s^2 / 100 deg/s^2), and max are highlighted as bold lines.

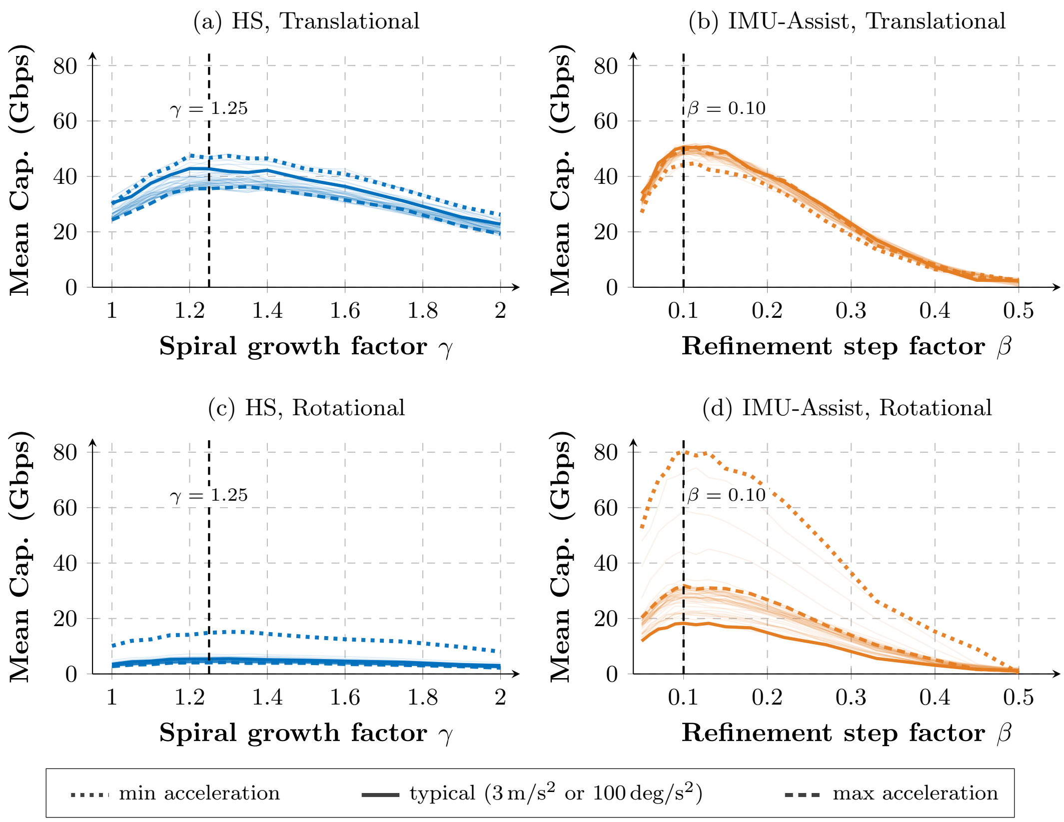

Figure 1: Baseline parameter sensitivity ($d_{\text{link}} = 3$ m, Ideal Mechanical) under Translational (a,b) and Rotational (c,d) scenarios, comparing Hierarchical Search (a,c) and IMU-Assist (b,d). Bold lines mark the min, typical, and max accelerations.

The chosen $\gamma = 1.25$ and $\beta = 0.10$ (Fig. 1) sit within $\pm 10\%$ of the peak of their respective sensitivity curves; outside this band, both baselines lose capacity, consistent with the trade-offs discussed in *Methods*. The chosen $\eta = 0.36$ (Fig. 2, left axis) is on a wide plateau (mean capacity within $\pm 5\%$ of the maximum across $\eta \in [0.20, 0.50]$). Even at the baselines' optima (HS at $\gamma = 1.25$: 42.8 Gbps; IMU-Assist at $\beta = 0.10$: 50.5 Gbps), JIT exceeds them by at least 26 Gbps at typical translational acceleration (76.6 Gbps at $\eta = 0.36$), so the gain reflects

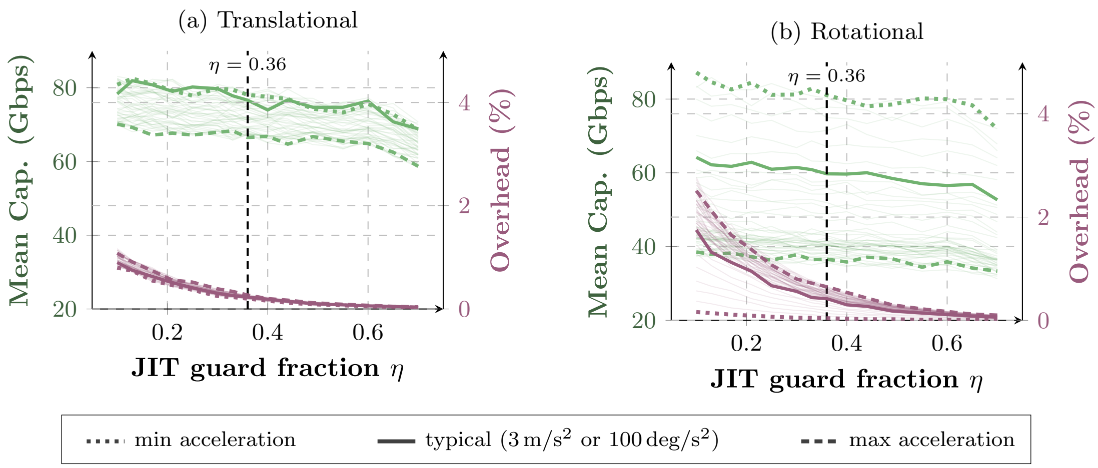

Figure 2: JIT guard fraction η sensitivity ($d_{\text{link}} = 3$ m, Ideal Mechanical) under (a) Translational and (b) Rotational scenarios. Bold lines mark the min, typical, and max accelerations.

the coordination mechanism rather than parameter tuning. The acceleration, distance, and beamwidth sweeps in *Results* show the same ordering at every operating point away from the typical scenario.

Analyze robustness of the trigger under long-term IMU drift without zero-velocity updates, and explain the effect of the 10 ms IMU sampling period on sub-millisecond control reaction.

Both points are addressed by new content in *Methods/JIT Beam Realignment*, with Fig. 3 providing the empirical sweep. JIT's residual position drift over a single prediction cycle is bounded to under 1 nm, far below θ_{HPBW}, and opportunistic ZUPT handles longer sessions. The new manuscript text is reproduced here for the Reviewer's convenience.

> Two design choices keep the residual drift small. The trigger condition uses the Link Deviation *velocity* rather than an absolute position (single vs double accelerometer integration), and the control frame transmits only the displacement change since the last transmission, with the accumulator reset at each transmission so cumulative drift cannot grow with session length. Over the AP's per-prediction window $T_{\text{react}} \approx 15\,\mu\text{s}$, the resulting position-prediction drift is bounded by $\sigma_p = \sigma_{a,n}\sqrt{T_{\text{react}}^3/3} \lesssim 1\,\text{nm}$ for the noise density in Table 2, an angular drift of order 10^{-8} degrees at $d_{\text{link}} = 3\,\text{m}$ and far below θ_{HPBW}. For multi-hour or multi-day operation, the ZUPT can be invoked opportunistically whenever the bias-corrected $\|\dot{\boldsymbol{\delta}}\| \approx 0$ persists for more than a few IMU samples (e.g., the device resting on a desk, charging, or in a pocket), re-estimating $\mathbf{b}_a, \mathbf{b}_\omega$ to within the sensor noise floor. For sustained continuous-mobility scenarios, complementary bias-tracking from the inertial-navigation literature such as learning-based dead-reckoning [13], deep-learning drift compensation [14], or zero-velocity detection [15] would augment JIT's calibration as future work.

The 10 ms IMU period gates only how often Eq. (7) is evaluated; the μs-scale control-frame, processing, and steering chain runs on its own clocks. The corresponding sentence in *Methods* is repeated below.

The IMU sampling period (10 ms in this evaluation; commodity ≥1 kHz IMUs such as the Bosch BMI270 are widely available) gates only the rate at which Eq. (7) is evaluated; once a trigger fires, the control-frame transmission, AP processing, and steering actuation all proceed on their own clocks, decoupled from the IMU sample interval.

Fig. 3 confirms this quantitatively: JIT mean capacity stays within 3% of its maximum across $T_{\mathrm{IMU}} \in [1, 15]$ ms and degrades smoothly thereafter, while IMU-Assist degrades steadily with T_{IMU} across the whole range. IMU-Assist runs an iterative refinement spiral after RF failure, updating its steering direction at each spiral step from the latest IMU sample; with coarser T_{IMU}, successive steps reuse near-identical samples and the spiral makes little progress per iteration. JIT, by contrast, fires its trigger pre-failure with the $\eta\,\theta_{\mathrm{HPBW}} \approx 4.7°$ safety margin in Eq. (7), so the lag from coarser T_{IMU} is absorbed by that margin. Human-scale UE motion evolves on millisecond-to-second timescales, so the IMU period in the tens of ms is not comparable to the motion itself, and JIT's gain comes from realigning in the predicted *direction* at the right moment rather than from raw reaction speed.

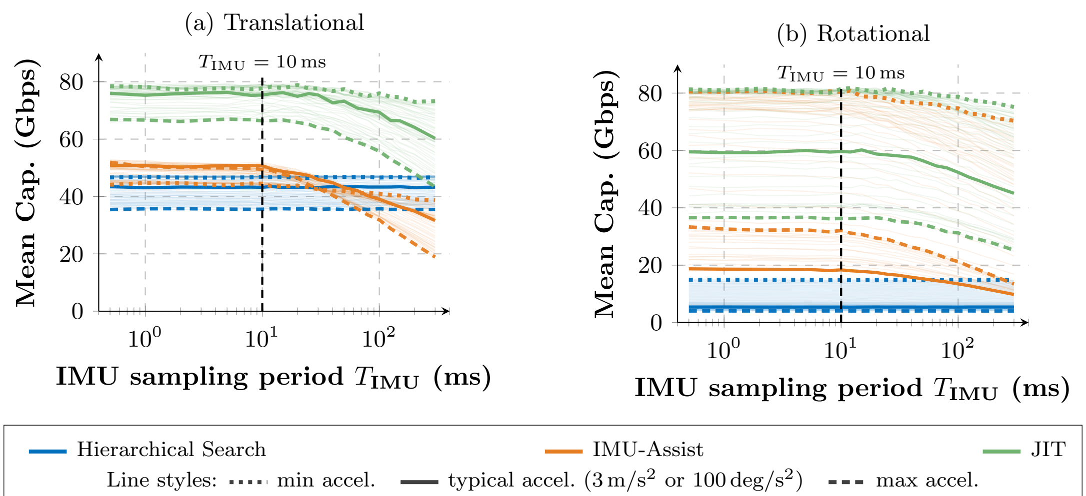

Figure 3: Mean capacity vs. IMU sampling period T_{IMU}, Ideal Mechanical steering ($d_{\mathrm{link}} = 3$ m). (a) Translational; (b) Rotational. Bold lines mark the min, typical, and max accelerations; vertical dashed lines mark the Table 2 baseline $T_{\mathrm{IMU}} = 10$ ms.

If the SNR falls below 9.59 dB the AP cannot decode the control frame and the system falls back to reactive recovery via Hierarchical Search, so the IMU prediction and the RF fallback are complementary rather than dependent.

Explain how the edge of coverage is defined and whether it adapts to UE speed or rotation. Also describe how the UE detects approaching EoC to transmit IMU data.

A new *Trigger Boundary and Edge of Coverage* subsubsection has been added at the head of *Methods/JIT Beam Realignment*. The EoC is the $\pm\theta_{\mathrm{HPBW}}/2$ off-boresight angle at which the antenna gain has dropped by 3 dB; JIT places its trigger boundary inside the EoC at the offset where the projected displacement over T_{react} equals $\eta\,\theta_{\mathrm{HPBW}}$, and the boundary contracts toward boresight as motion speed grows. We have duplicated the full subsubsection here so the Reviewer can read it in one place.

> The EoC is the angle off the AP's beam axis at which the antenna gain has dropped by 3 dB, i.e., $\theta_{\mathrm{HPBW}}/2$ off boresight. Crossing the EoC corresponds to a worst-case 6 dB round-trip gain loss, particularly punishing in sub-THz where narrow, high-gain beams leave little SNR margin, and can directly cause link failure. Because reaching the EoC under mobility leaves no time for control signaling, JIT instead places a *trigger boundary* inside the EoC, at the offset where the projected displacement during the system reaction time T_{react} would reach $\eta \cdot \theta_{\mathrm{HPBW}}$. The boundary adapts to motion: as the UE's Link Deviation Velocity $\|\dot{\boldsymbol{\delta}}\|$ grows, the projected displacement reaches $\eta \cdot \theta_{\mathrm{HPBW}}$ at a smaller angular offset from boresight, so the trigger fires earlier, farther inside the EoC. When motion slows, the boundary moves outward toward the EoC, suppressing unnecessary control frames. For AP-stationary scenarios (e.g., a wall-mounted access point serving mobile UEs), detection of approaching EoC is performed locally at the UE: at every IMU sample, the UE evaluates Eq. (7) using its body-frame kinematics and the known θ_{HPBW} of its own antenna, requiring no AP-side feedback. In future work, mobile-AP cases (e.g., UAV-borne access points) would extend the protocol so the AP also shares its kinematic state via the same control frame.

If the IMU-based prediction underestimates the crossing and the SNR drops below the control-frame decoding threshold (9.59 dB), the AP falls back to the Hierarchical Search baseline already characterized in the main paper, so the link is never lost silently.

We sincerely thank the Editor for these comments, which together have produced a clearer and more rigorous presentation of JIT's design rationale, parameter sensitivity, and operating range.

Reviewer 1

> The novelty is meaningful, but its boundary relative to prior sensor-assisted/predictive beam management remains under-specified. ... A clearer related-work positioning would strengthen the innovation claim.

We agree, and have expanded the related-work positioning in the manuscript and added Table 1 placing JIT against six families of prior beam-management work. What is new in JIT is the combination of UE-side IMU motion sensing, a pre-failure trigger that anticipates the next beam-edge crossing, and coordinated AP/UE realignment delivered via a single in-band sub-THz control frame; to our knowledge, no prior work combines all three within a single sub-THz protocol with hardware-in-the-loop validation. The per-family limitations are as follows: 3GPP NR Beam Failure Recovery [1, 2] sends a discrete PRACH-based recovery message only after the MAC entity has counted N consecutive Beam Failure Instances at L1; RF-based proactive failure prediction [3, 4] forecasts a failure from RF time-series or sub-6 GHz features and uses no motion sensing; AP-side multimodal sensing [5, 6, 7] requires LiDAR, radar, vision, or GPS hardware at every AP; UE-side orientation-based work [8, 9] uses orientation primarily for codebook selection rather than to anticipate the next edge crossing; continuous IMU streaming [10] exchanges IMU data on every tracking interval over an out-of-band channel; and bilateral pilot-driven coordination [11, 12] incurs overhead regardless of link risk. JIT sends one short in-band frame only when a crossing is imminent, for an overhead of less than 0.4%.

The two reactive baselines (Hierarchical Search and IMU-Assist) remain in the experimental comparison because they share JIT's protocol-level scope; Static and Perfect Alignment provide lower and upper reference bounds. The other approaches in Table 1 operate at a different protocol scope (standardized post-failure recovery, AP-side LiDAR/GPS, periodic bilateral pilots), and a like-for-like experimental comparison against any of them is a separate study. The *Discussion* future-work paragraph now lists this explicitly; the new passage is shown below for reference.

> ... the validation scope can broaden toward (i) multi-AP scheduling with handover, (ii) non-line-of-sight channel models with multipath and transient blockage events, (iii) HIL trials in which both nodes translate through azimuth and altitude, (iv) sustained continuous-mobility scenarios with complementary bias-tracking mechanisms beyond the present initial-event focus, and (v) larger-scale measurement campaigns spanning realistic indoor and outdoor environments.

> Insufficient tuning transparency, sensitivity analysis, and ablation make the source of JIT's gains hard to interpret...

We have added two parameter-tuning figures (Fig. 1 sweeps γ and β; Fig. 2 sweeps η across both motion classes). Each parameter sweep uses the same translational (0.5–20 m/s^2) and rotational (0.5–700 deg/s^2) peak acceleration grids used throughout the main paper, with the same $n = 5{,}000$ Monte-Carlo runs per data point. The full sweep is drawn as faint background lines; the min, typical (3 m/s^2 / 100 deg/s^2), and max accelerations are highlighted as bold lines. The chosen baseline parameters ($\gamma = 1.25$, $\beta = 0.10$) lie within $\pm 10\%$ of the peak at every acceleration shown, confirming that the choice is not specific to one acceleration. JIT's mean capacity stays within $\pm 5\%$ of its maximum across the plateau $\eta \in [0.20, 0.50]$ for every acceleration. Even at the baselines' optima (HS at $\gamma = 1.25$: 42.8 Gbps; IMU-Assist at $\beta = 0.10$: 50.5 Gbps), JIT at $\eta = 0.36$ leads by at least 26 Gbps (76.6 Gbps) at typical translational acceleration, so the gain reflects the coordination mechanism rather than parameter tuning.

The companion timescale sweep (Fig. 3, T_{IMU}) is addressed in our response to the next remark, and confirms that JIT capacity stays within 3% of its maximum for $T_{\text{IMU}} \leq 15\,\text{ms}$ (commodity IMUs operate at $\geq 1\,\text{kHz}$, well inside this range).

> Several key engineering assumptions are under-analyzed.... However, it remains unclear how robust the proactive trigger is under longer continuous operation without reliable zero-velocity intervals.... The effect of this sensing timescale on trigger freshness and prediction accuracy is not adequately analyzed.

Two design choices bound the residual IMU drift on both per-prediction and longer timescales. First, the trigger uses the Link Deviation *velocity* (single accelerometer integration), not absolute position (double integration); over the AP's per-prediction window $T_{\text{react}} \approx 15\,\mu\text{s}$, the residual position drift is $\sigma_p = \sigma_{a,n}\sqrt{T_{\text{react}}^3/3} \lesssim 1\,\text{nm}$ for the noise density in Table 2, an angular drift of order 10^{-8} degrees at $d_{\text{link}} = 3\,\text{m}$ and far below θ_{HPBW}. This bound has no dependence on session length. Second, the JIT control frame transmits only the displacement change since the last transmission, with the on-UE accumulator reset at each transmission, so cumulative drift inside the frame payload is bounded by a single inter-transmission interval rather than the full session duration. Across multi-hour or multi-day operation, the initialization ZUPT can be invoked opportunistically: any interval during which the bias-corrected $\|\dot{\boldsymbol{\delta}}\| \approx 0$ persists for more than a few IMU samples (device on a desk, charging, or in a pocket between motion bursts) is an implicit zero-velocity interval and can be used to re-estimate $\mathbf{b}_a, \mathbf{b}_\omega$ to within the sensor noise floor without user action.

The $10\,\text{ms}$ IMU period gates only how often Eq. (7) is evaluated; once a trigger fires, the control-frame transmission ($T_{\text{frame}} \approx 5\,\mu\text{s}$), the AP processing ($T_{\text{proc}} = 10\,\mu\text{s}$), and the steering actuation proceed on their own clocks. JIT capacity stays within 3% of its maximum across $T_{\text{IMU}} \in [1, 15]\,\text{ms}$ (Fig. 3), so T_{IMU} is not the control-reaction bottleneck across the commodity range.

> The current validation scope remains simplified (single AP–single UE, predominantly LoS/FSPL-based modeling, and HIL constrained to the azimuth plane), which limits the external validity of some design insights, such as the robustness and low-overhead operation of JIT.

We agree that this is first-order validation, and have made the scoping explicit in the revised *Statistics and Reproducibility* subsection. The single-link, predominantly LoS configuration isolates the beam-management mechanism from the confounding effects of multi-AP, NLoS, or larger-scale settings.

The dual-axis rotary stages support full 3D antenna pointing in both azimuth and altitude; the HIL antenna pointing is therefore 3D. The constraint is on the linear translation of the RF front-end mounts: each front-end (PSG + AWG + upconverter on the Tx side, downconverter + oscilloscope on the Rx side) is too heavy to lift in the vertical direction, so the linear translation is restricted to the laboratory-floor azimuth plane. This is now stated explicitly in *Statistics and Reproducibility*. For the Reviewer's convenience, we have included the new sentence below.

> The dual-axis rotary stages support full 3D antenna pointing in both azimuth and altitude; the linear translation of the RF front-end mounts, however, was constrained to the azimuth plane by the weight of the RF equipment.

JIT's robustness under 3D motion follows from the trigger condition (Eq. 7), not from the HIL geometry: Eq. (7) is evaluated in 3D world coordinates, and the simulator exercises 3D motion in the rotational-acceleration, angular-displacement, and combined-motion sweeps (Figs. 4–8). The low-overhead claim ($O_{\text{sig}} < 0.4\%$) likewise follows from the event-driven trigger and the IEEE 802.15.3d THz-SC frame budget, not from the HIL constraint.

The five validation-scope extensions are listed in the *Discussion* future-work paragraph, reproduced in our response to Reviewer 1, Comment 1 above.

> The "Link Deviation Velocity" $\dot{\boldsymbol{\delta}}$ in equation (6) combines UE angular velocity and LoS-perpendicular translational velocity divided by distance. It would benefit from a clearer explanation of how these quantities are represented and combined in the 3D coordinate frame, especially whether both terms are projected onto the same beam-deviation angular coordinates.

We agree, and have made the coordinate convention explicit. Both terms in Eq. (6) are expressed in the AP-anchored world coordinate frame via the UE orientation quaternion $\mathbf{q}_{\text{body}}$, so rotational and LoS-perpendicular translational contributions sum consistently as a single 3D angular rate. Two clarifying sentences have been added immediately after Eq. (6) in *Methods/JIT Beam Realignment*, and a complete notation summary has been added as a new Table 3 at the head of *Methods*. The two new sentences are duplicated here.

> Both terms in Eq. (6) are expressed in the AP-anchored world coordinate frame: the UE's body-frame angular velocity $\boldsymbol{\omega}_{\text{UE}}$ and translational velocity perpendicular to the LoS, $\mathbf{v}_{\perp}$, are transformed via the UE's orientation quaternion $\mathbf{q}_{\text{body}}$ (the same quaternion that contributes to $\mathbf{q}_{\text{eff}}$ in the antenna pattern evaluation discussed below). This common frame ensures that rotational and LoS-perpendicular translational contributions to beam deviation are summed consistently as a single 3D angular rate.

> In the subsection of JIT Beam Realignment, the derivation of the filtered acceleration trend $\bar{\mathbf{a}}_{\text{predict}}$ from EWMA needs further explanation.

The revised *Methods/JIT Beam Realignment* subsection now states the recurrence explicitly as

$$\bar{\mathbf{a}}_{\text{predict},k} = \alpha_{\text{short}}\,\mathbf{a}_k + (1-\alpha_{\text{short}})\,\bar{\mathbf{a}}_{\text{predict},k-1}, \tag{1}$$

with $\alpha_{\text{short}} = 0.90$ as listed in Table 2. This low-pass cutoff suppresses high-frequency IMU sensor noise while preserving the underlying acceleration trend, which the AP uses to extrapolate the UE's position at steering completion.

> The Perlin smootherstep function in equation (8) mathematically forces velocity and acceleration to zero at keyframe boundaries. While this is reasonable for generating smooth trajectories, the authors should explain why this is an adequate approximation for the considered mobility scenarios.

The clarification has been added immediately after the smootherstep equation in *Methods*. The zero-velocity boundary models pause-resume mobility, which is the hardest case for predictive beam management; reported performance therefore lower-bounds what is achievable on sustained-motion variants. The new passage from *Methods* is copied below.

> The zero-velocity boundary condition models pause-resume mobility patterns such as walking with intermediate stops or beginning and ending a turn from rest, which are the primary motion classes considered in this study. Pause-resume motion is also the hardest case for predictive beam management: the velocity profile is unknown at the onset of motion, acceleration changes rapidly, and the system has the

> least time to react. Sustained-motion variants (e.g., walking at a steady speed) are easier to predict from a brief observation window, so the performance reported here lower-bounds what is achievable on those variants. Both the software simulator and the HIL testbed support trajectories without a stationary boundary (e.g., continuous walking through multiple checkpoints) by chaining smootherstep-interpolated segments whose endpoints exchange non-zero velocities, preserving slot-by-slot C^1 continuity where required.

> Equations (12) and (13) explicitly model the E-plane pattern of the WR-6.5 horn, but it is not clear whether the 3D kinematic simulation assumes a rotationally symmetric pattern based on the E-plane.

The Reviewer's interpretation is correct. The assumption is now stated immediately after the Fresnel-arguments equation in *Methods*. The new text is duplicated here.

> For 3D evaluation, the radiation pattern is treated as rotationally symmetric about the boresight: the gain $G(\theta)$ depends only on the off-axis angle θ obtained by transforming the 3D LoS vector into the antenna's local frame via $\mathbf{q}_{\text{eff}}$, and the E-plane formula above is applied across all azimuth angles around boresight as a rotationally symmetric approximation. The HIL measurements cross-check this approximation against the physical horn and confirm that it is sufficient for this evaluation.

The HIL error bands (translation: Mean Absolute Percentage Error (MAPE) 1.80–3.13%, Normalised Root-Mean-Square Error (NRMSE) 2.99–3.82%; rotation: MAPE 3.88–4.68% for JIT and IMU-Assist, 11.66% for Hierarchical Search due to its low absolute capacity) are reported in *Results*.

We sincerely thank Reviewer 1 for the careful read and thoughtful comments, which together have led to a clearer and more rigorous presentation of the methods and evaluation scope.

Reviewer 2

> The setting of the "trigger boundary" in the JIT method is not well explained. Specifically, how is EoC determined, and will it be adjusted based on the user's speed or rotation?

The new *Trigger Boundary and Edge of Coverage* subsubsection in *Methods/JIT Beam Realignment*, reproduced in our response to the Editor's Comment 4 above, addresses both questions. The EoC is the $\pm\,\theta_{\mathrm{HPBW}}/2$ off-boresight angle at which the antenna gain has dropped by 3 dB, and JIT places an inner *trigger boundary* at the offset where the projected angular displacement over T_{react} equals $\eta\,\theta_{\mathrm{HPBW}}$. The boundary adapts to motion through $\dot{\boldsymbol{\delta}}$: faster motion (larger $\|\dot{\boldsymbol{\delta}}\|$) contracts the boundary toward boresight and fires the trigger earlier; slower motion expands it toward the EoC and suppresses unnecessary control frames. Under pure rotation $\mathbf{v}_{\perp} \approx 0$ and the trigger fires off the rotational term alone; under pure translation $\boldsymbol{\omega}_{\mathrm{UE}} \approx 0$ and it fires off the translational term alone.

> Since the user is within the beam coverage, how can it determine that it is approaching the EoC in order to transmit its IMU message?

Detection is performed locally at the UE, without AP-side RF feedback. At every IMU sample the UE applies the ZUPT-derived biases to obtain $\mathbf{a}_{\mathrm{corr}}, \boldsymbol{\omega}_{\mathrm{corr}}$, transforms its body-frame velocities into the AP-anchored world frame via $\mathbf{q}_{\mathrm{body}}$, computes $\dot{\boldsymbol{\delta}}$ using the last known-good d_{link} from the RF link, and tests $\|\dot{\boldsymbol{\delta}}\| \cdot T_{\mathrm{react}} > \eta\,\theta_{\mathrm{HPBW}}$. All four required quantities are available on the UE side (IMU readings, orientation quaternion, last known link distance, and the antenna's θ_{HPBW}), so the UE can predict the future EoC crossing from its own kinematic state and transmit a control frame before the link is lost. The trigger uses velocity rather than position to avoid double-integration drift; opportunistic ZUPT bounds the residual bias; and the SNR-driven fallback at 9.59 dB reverts the system to the Hierarchical Search baseline if the IMU-based prediction underestimates the crossing, so the link is never lost silently. For mobile-AP cases (e.g., UAV-borne access points), the same control frame is the natural carrier for the AP's kinematic state; this extension is on the future-work list.

> System configurations and KPI definitions should be introduced before the results section to enhance the readability of the Results. Please restructure the paper for better clarity.

The opening of *Results* has been restructured: the four reference schemes, the three steering models, and the four KPIs are now grouped under a new labelled subsection at the head of *Results*, with full derivations left in *Methods/Evaluation Metrics*. We have included the new subsection heading and its opening below.

> **Experimental Setup and Performance Metrics.** We benchmark JIT against four reference schemes... We evaluate each scheme across four key performance indicators (KPIs)...: *Mean Capacity* ($\bar{C}$)...; *Capacity Coefficient of Variation (CoV)*...; *Link Availability* (A_{link})...; and *Signaling Overhead* (O_{sig})...

> On page 3, when referring to the "state-of-the-art beam recovery approach (without IMU assistance)," it is unclear which specific work this refers to. Please provide a reference and more details about this approach.

The intended comparator is the IEEE 802.11ad / 3GPP NR-style hierarchical sequential beam search. In the revised *Results* preamble, the phrase "state-of-the-art beam recovery approach (without IMU assistance)" has been replaced with "state-of-the-art Hierarchical Search beam recovery approach (without IMU assistance) [16], consistent with the SSB-based beam search defined for 3GPP NR [1, 2]". The same citations have been added in *Methods/Hierarchical Search* where the spiral pattern is introduced. For the IMU-assisted baseline, the mmWave IMU-assisted beam-tracking literature [17, 10, 18] is now cited in *Methods/IMU-Assisted Reactive Search*.

> In Figure 2, the JIT method shows trends that seem to contradict the IMU-assisted approach at high acceleration. What limitations in the JIT method explain this divergence from IMU-assisted performance? In contrast, in Figure 4, why does JIT match the IMU trend in the electronic model? Please explain the differences and the reasons behind this phenomenon.

Figs. 2 and 4 sweep different motion types; the JIT-vs-IMU-Assist gap depends on whether AP-side action is needed.

A translational displacement moves the UE in the AP's world frame, so the AP boresight that was correct before the motion no longer points at the UE. IMU-Assist updates only the UE side, so the AP error persists; as the acceleration grows the beam residence time shrinks and the AP error accumulates, while JIT updates both nodes via the in-band control frame. The IMU-Assist–JIT gap therefore widens with acceleration in Fig. 2.

A pure rotation of the UE does not change its position in the AP's frame; the AP boresight remains correct and only the UE-side pointing has to be updated. Under Electronic steering the UE-side update completes within microseconds, so IMU-Assist tracks even $700\,\mathrm{deg/s^2}$ and the relative gap to JIT collapses (Fig. 4(c)). JIT retains an absolute lead of 19 Gbps at $700\,\mathrm{deg/s^2}$, attributable to the IMU-Assist overshoot at low acceleration described in *Methods/IMU-Assisted Reactive Search*, which the JIT predictive trigger avoids by construction.

The same mechanism applies to Fig. 5 under Ideal Mechanical steering: the slower UE-side hardware cannot complete a counter-rotation within the residence time, and only JIT sustains the link across the full rotational range. The *Discussion* of the revised manuscript notes this distinction: translational motion always requires AP action; rotation requires it only when UE-side steering is too slow.

> In Figure 3, after beam alignment, why is there a power deviation compared to the "perfect alignment" case? What does this deviation represent? Why does JIT's trajectory information not seem to assist in perfectly aligning the beam after realignment?

The residual is a consequence of JIT's event-driven design rather than a deficiency of the realignment itself. A new paragraph in *Results* states this mechanism explicitly and ties it back to Eq. (7). The new paragraph is reproduced below for the Reviewer's convenience.

> A small residual gap between JIT and the Perfect Alignment reference is visible in the steady-state portion of Fig. 3. By design, JIT does not continuously chase the maximum SNR; it triggers realignment only when the projected angular displacement is about to cross $\eta \cdot \theta_{\mathrm{HPBW}} = 0.36 \cdot 13^\circ \approx 4.7^\circ$ (Eq. 7). Between triggers, the UE may sit slightly off-axis without correction, while the Perfect Alignment reference assumes zero misalignment at every instant.

We sincerely thank Reviewer 2 for the close reading of the figures, which has prompted us to articulate JIT's event-triggered behavior and the figure-to-figure consistency more clearly than in the original submission.

We sincerely thank the Editor and the Reviewers for the thoughtful and constructive feedback, which has substantially improved the clarity and rigor of the manuscript. We hope the revisions adequately address every concern raised, and we look forward to your further consideration.

Sincerely,

Sergey Petrushkevich, Vitaly Petrov, and Josep M. Jornet

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